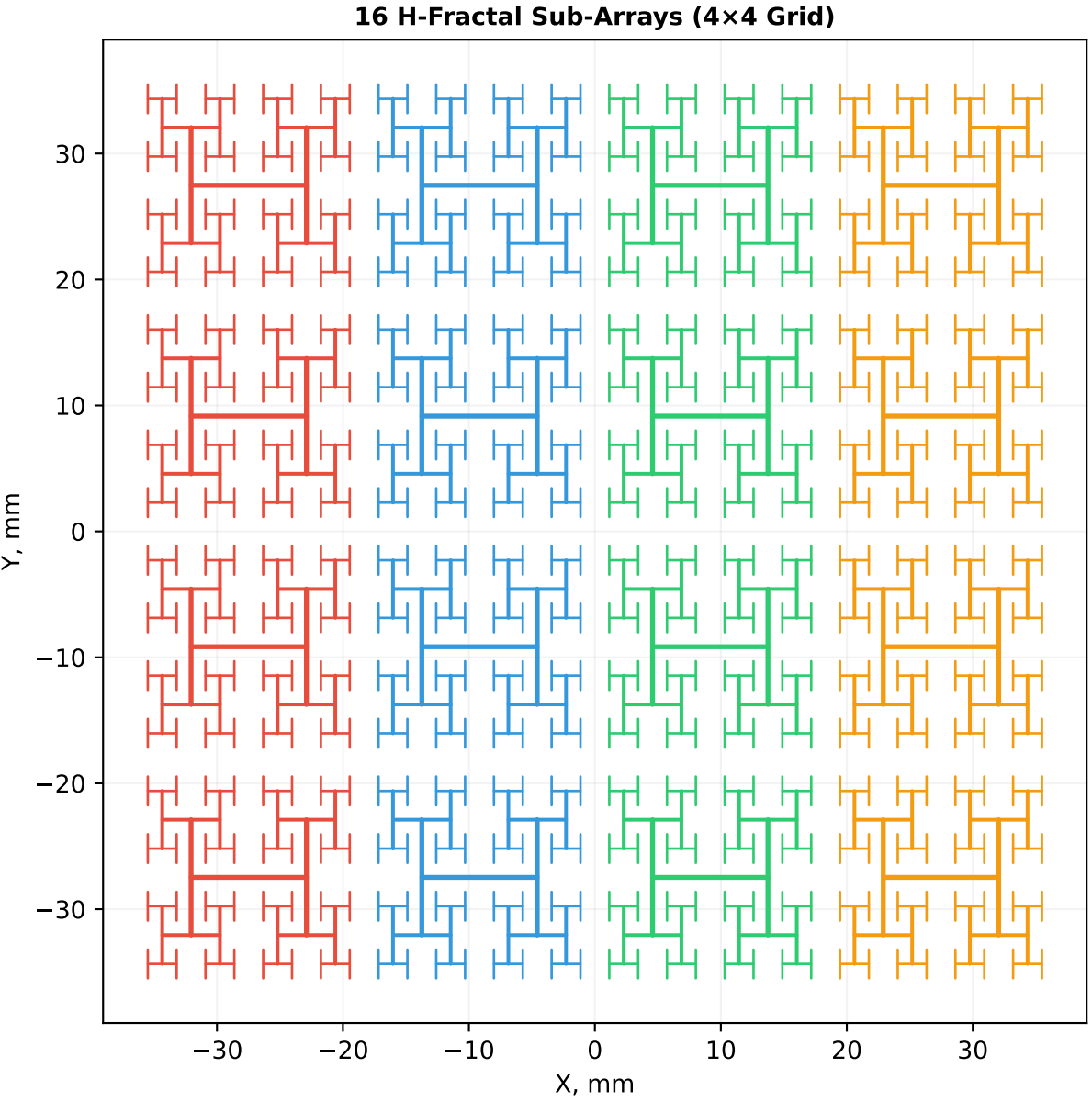
Fractal 32×16 — 16-Channel Simulation Report

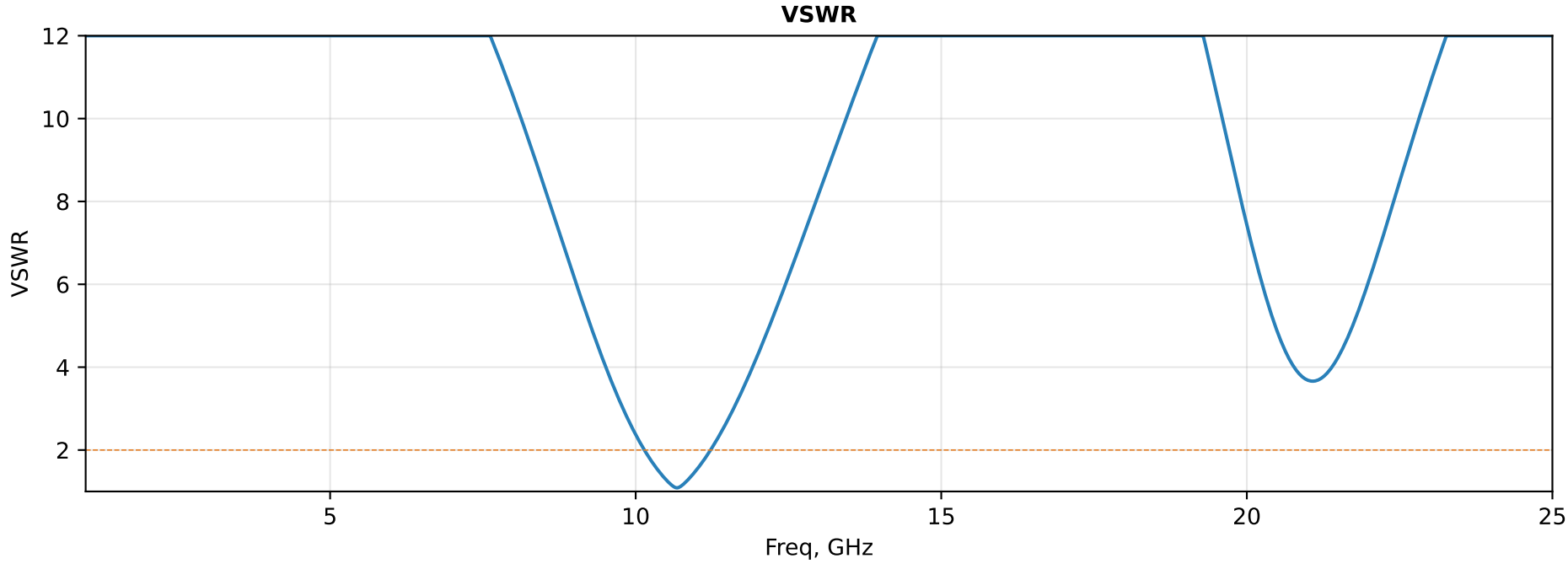
H-Fractal Scanning Phased Array · 10.5 GHz · 4× ADAR1000



Antenna:	Fractal 32×16
Topology:	H-Fractal 4×4
Channels:	16
Frequency:	10.5 GHz
λ_0:	28.55 mm
λ_g:	18.32 mm
Substrate:	RO4350B
Sub-array:	9.16 mm × depth 3
Spacing:	18.3 mm (0.64λ)
Beamformer:	4× ADAR1000
Board:	90 × 120 mm

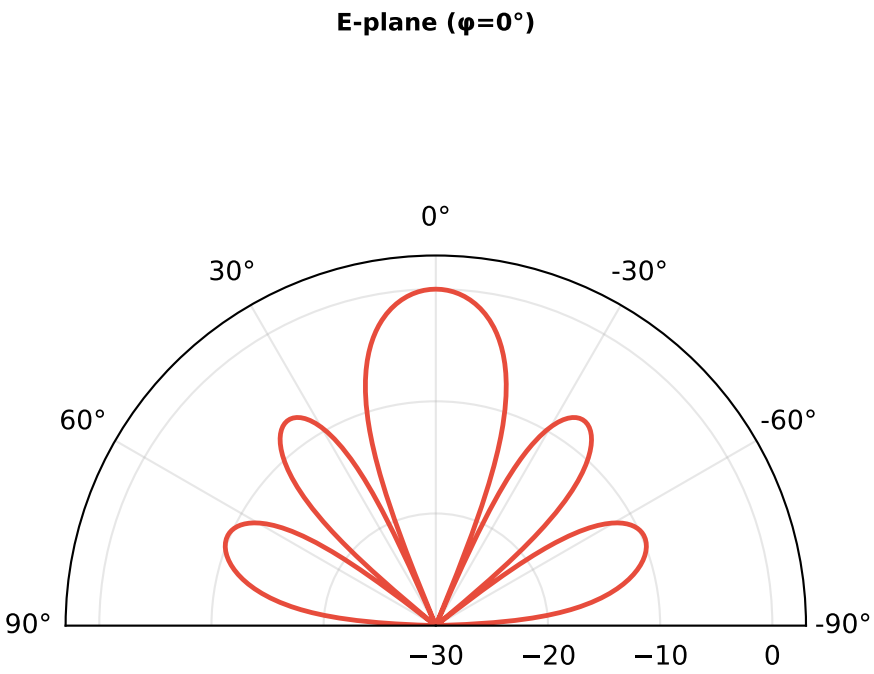
S_{11} and VSWR (1–25 GHz)

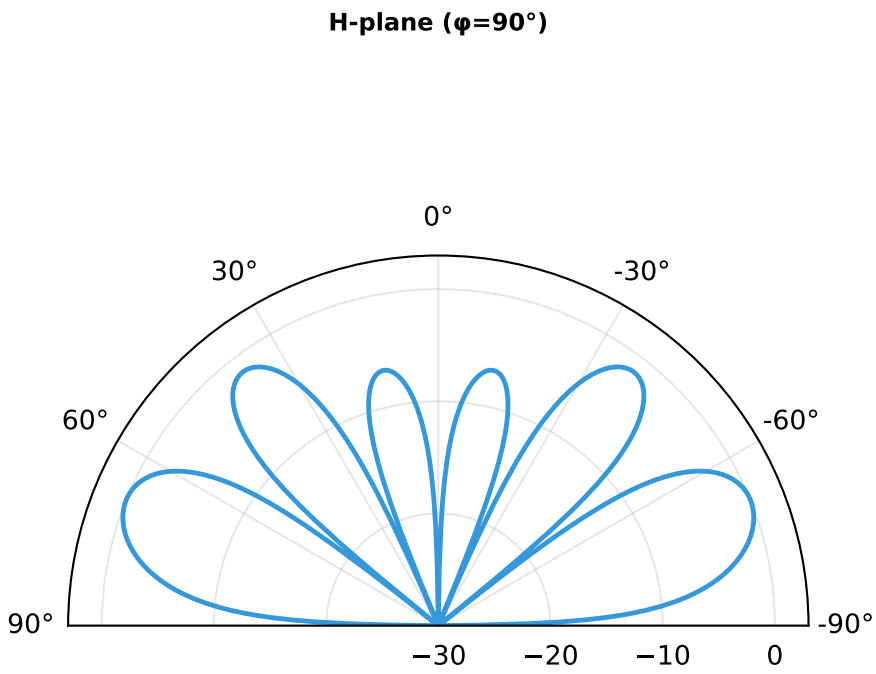
16-element array · multi-band fractal resonances



Radiation Pattern at 10.5 GHz (broadside)

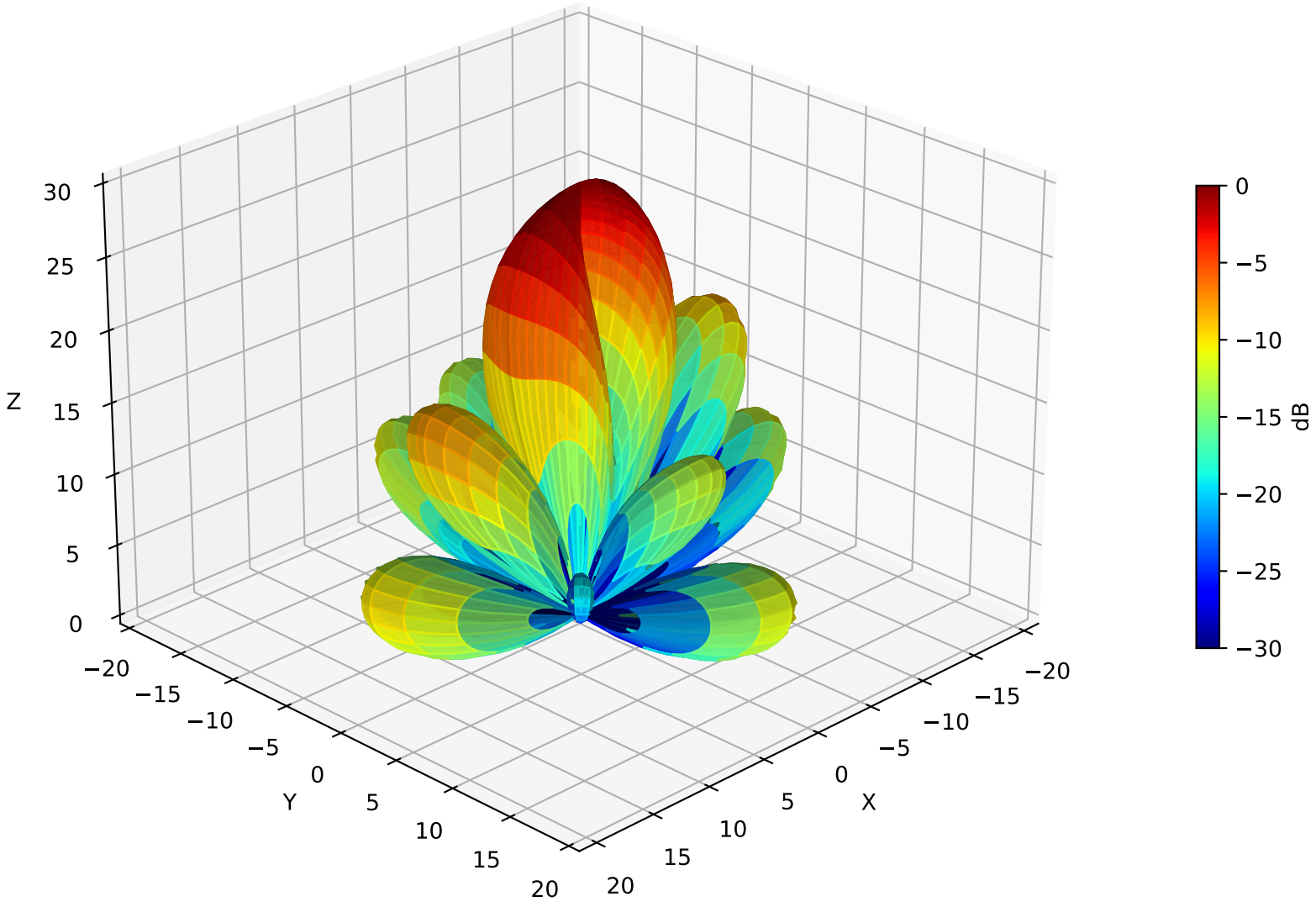
4×4 array · normalized to peak





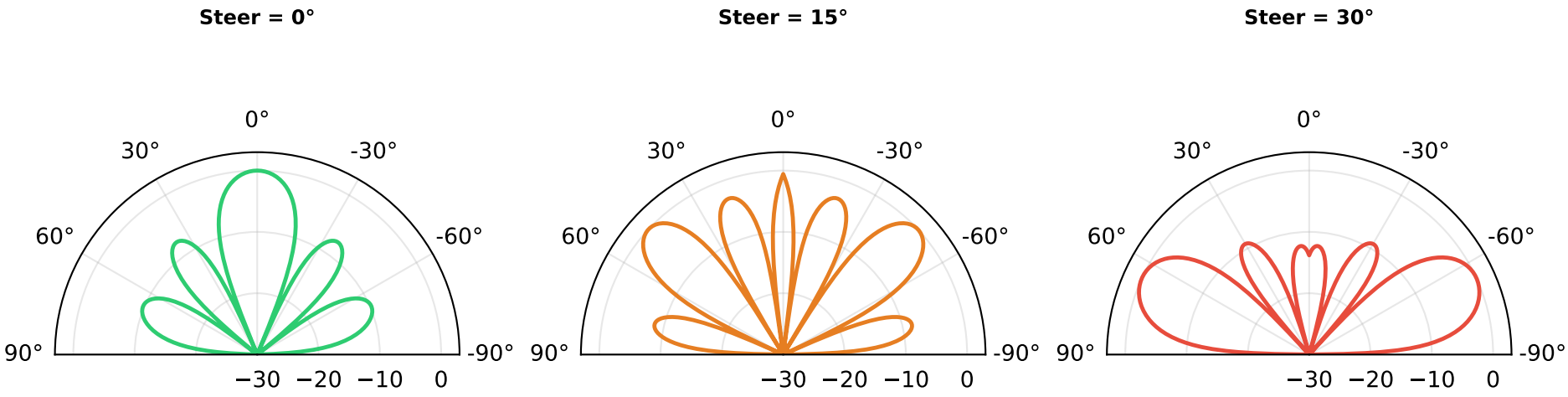
3D Radiation Pattern (upper hemisphere)

16-element array · broadside beam



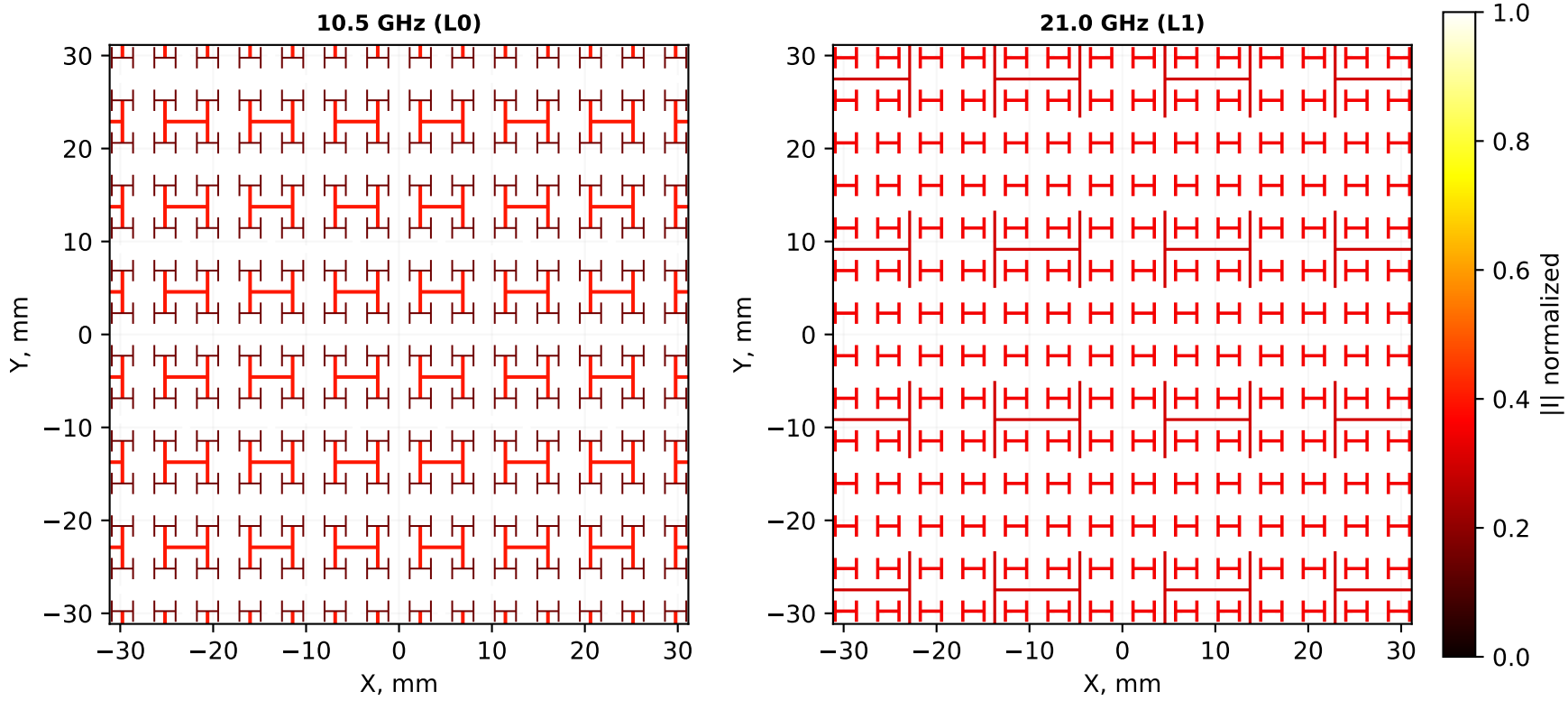
Beam Steering with 4× ADAR1000 (E-plane)

Column-phase gradient applied to 4×4 array



Current Distribution on 4×4 H-Fractal Grid

Color ∝ current magnitude per fractal level



Simulation Results Summary

Fractal 32×16 · 16-channel phased array

Parameter	Value	Unit
Design freq	10.5	GHz
S_{11} @ f_0	-18.3	dB
VSWR	1.28	
Z_in	49.3+j12.2	Ω
-10dB BW	1015	MHz
Directivity	18.6	dBi
Efficiency	65	%
Realized gain	16.8	dBi
Array	4×4 (16 sub-fractals)	
Beamformer	4× ADAR1000	
Scan range	±30	deg
L0 resonance	10.5	GHz
L1 resonance	21.0	GHz
L2 resonance	42.0	GHz